

PRACTICUM WORKSHEET: THE AMDAHL REALITY GAP

Empirical Speedup Fitting, Micro-Architectural Bottlenecks, and Loop Scheduling in OpenMP

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Institutional Email / Git URL: __230107088@sdu.edu.kz/ a1zzyBd/CSS410Lab1 _____	Practicum Date & Time (Slot): __monday 08:30__
Laptop Model & OS (Linux/macOS/Win): __HUAWED-14	CPU Model & Generation: __AMD Ryzen 7 4800H with Radeon Graphics__
Physical Cores (P/E breakdown if hybrid): __Physical cores: 8 Hybrid P/E: No (AMD, non-hybrid)	Logical SMT Threads (omp_get_max_threads): __16__
Target Workload Size N (Derived via Formula): __7098000__	L1 / L2 Cache Line Size (Typically 64 bytes): __L1:512KB,L2:4096 MB

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PS C:\Users\Huawei\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Huawei\AppData\Local\Python\pythoncore-3.14-64\python.exe d:/collatz_lab_full.py
detected logical CPU count: 16
thread counts to test: [1, 2, 4, 8, 16]

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PHASE 2: Sequential Baseline (3 runs, discard Run 1 as warmup)
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Run 1: time=569.0315s max_steps=704 checksum=747975164 (WARMUP - discarded)
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